PORTGEN

Dynamic Online Portfolio Generator

Project type: School software system documentation

Live URL: https://portgen-m1uh-if1e.onrender.com

Target audience: Students, freelancers, and job seekers

Author: PortGen project implementation

Date: 2026-10-20

Version: 0.1.0

PortGen is a Laravel application for building and publishing a personal portfolio from one structured form. This paper documents the implemented system, the evidence gathered from automated checks and expert inspection, and the Render deployment plan. The public source repository contains the application, design records, and HCI evidence. No classmates have taken part in the usability study yet.

1. System overview

1.1 Problem and solution

Students and early-career professionals often have information spread across documents and profile sites. A portfolio can be time-consuming to format and keep consistent. PortGen collects a person's introduction, education, work experience, skills, projects, and social links in one guided form. The person chooses a responsive presentation and receives a public portfolio URL. Account owners manage portfolios from their dashboard; guests receive a separate signed private link.

1.2 Objectives

1. Collect structured portfolio information in a guided, recoverable flow.
2. Render the same information with Simple, Modern, or Creative presentation styles.
3. Let account owners manage portfolios from a dashboard and let guests manage portfolios through a signed private link.
4. Support keyboard operation, responsive layouts, clear feedback, and a dark color scheme.
5. Keep database constraints, request validation, automated browser checks, and HCI evidence in the project repository.

1.3 Scope and limits

The implementation includes portfolio creation, template selection, public viewing, session-based registration and login, an account dashboard, signed guest-link management, portfolio claiming, editing, deletion with a ten-second undo window, friendly error pages, a demo portfolio, CI configuration, and Docker deployment configuration for Render. It does not include password recovery, image uploads, payments, analytics, or user research results. The public demo contains fictional information.

Laravel 11 was upgraded to Laravel 12.70.0 after its security-fix window ended; the upgraded dependency set currently passes `composer audit --locked`. The deployment branch now defines a free Render web service and a dedicated Render Free Postgres database for demonstration use. Render resources and live health checks are still pending. The database expires after 30 days, followed by a 14-day upgrade grace period before deletion. See section 8 for the release configuration and current status.

2. Users and needs

2.1 Personas and tasks

Table 1. Target users and primary needs

Persona	Situation and tasks	Likely needs
Student	Building an early-career profile with limited work history; adds education, skills, and coursework projects, selects a template, and shares the page.	Examples for unfamiliar fields, useful empty states, and a way to show projects without paid experience.
Freelancer	Presents services and selected client work; edits the portfolio as work changes and shares it with prospects.	Fast editing, a work-first layout, and a private owner link that can be saved.
Job seeker	Prepares a focused profile for applications; selects a readable layout and revises details.	Straightforward form, keyboard access, clear completion feedback, and predictable editing and deletion.

These are design personas and assumptions, not research findings. The main assumptions are that users can provide the information in one sitting if examples are available, that the public and private links are understandable, and that three initial templates provide enough variety. The study kit in `docs/usability-testing/` leaves participant outcomes open for classmates.

2.2 User-needs walkthrough

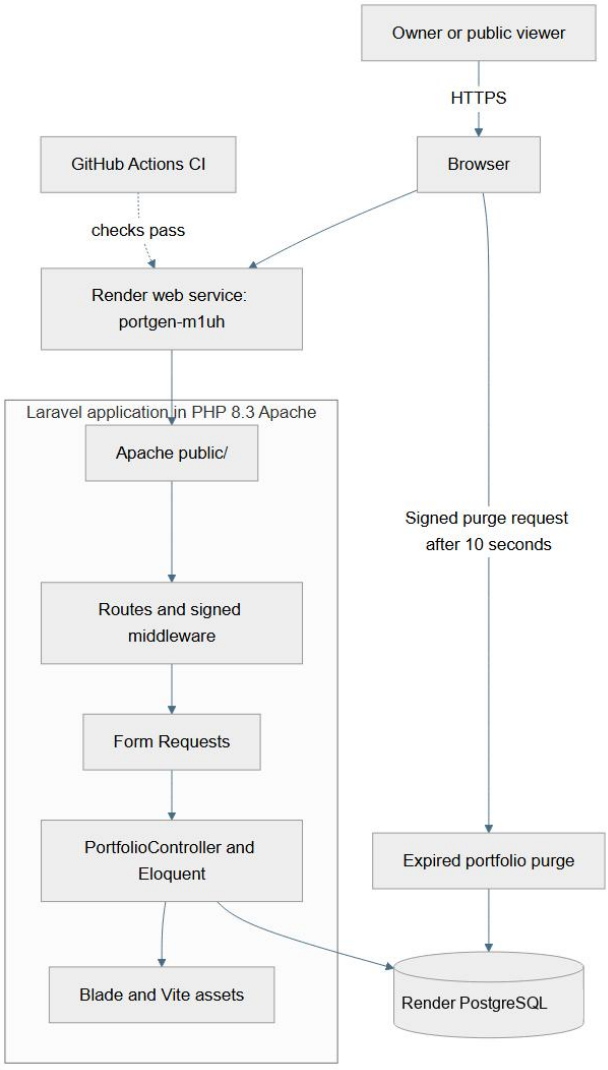
The student path permits empty education and work sections while requiring at least one skill. This supports users with limited employment history while ensuring that a portfolio has at least one concrete strength. The freelancer path supports repeatable project entries and an owner link for later changes. The job-seeker path uses familiar labels, inline validation, a completion stage, and a confirmation before deletion. These are expert walkthrough conclusions to validate with participants.

3. System architecture and technology stack

3.1 Architecture

The browser submits validated portfolio fields to Laravel. The controller and Eloquent model persist structured values in PostgreSQL JSONB columns. Blade renders the public page from the chosen template. Authenticated owners manage account-owned portfolios through their dashboard; guests use a signed private link. On Render, the browser's signed purge request and the Laravel scheduler remove a deleted record when its ten-second undo period ends.

Figure 1. Application architecture



The user-facing application is a modular Laravel monolith. The controller handles the portfolio workflow, Form Requests validate input, signed middleware protects owner routes, Blade templates render screens, and the database stores portfolio records. Render serves the app with Apache and supervises `schedule:work` alongside the web server.

3.2 Technology stack

Table 2. Implemented and planned technology

Layer	Technology	Use and status
Server framework	Laravel 12.70.0, PHP 8.3 target	MVC routing, validation, ORM, migrations, signed URLs, and scheduled commands. The local workstation runs PHP 8.5.
Views and styling	Blade, Tailwind CSS 4, Vite 8	Server-rendered views, design tokens, responsive styling, and production asset bundles.
Database	PostgreSQL 16 in CI and Render	The CI service is isolated. A dedicated Render Free Postgres database is configured for demonstration deployment and has a 30-day expiration.

Layer	Technology	Use and status
Browser checks	Playwright 1.64 and axe-core 4.14	Keyboard journey, responsive screenshots, touch target checks, and accessibility audits.
Container	Docker, PHP 8.3 Apache on Render	Multi-stage image configuration is committed; the Render image build runs in GitHub Actions.
Hosting	Render web service with Render Postgres	PostgreSQL stores portfolio data, sessions, and cache. The Free database is temporary demo storage.
CI	GitHub Actions	Pull requests and main run PHP, PostgreSQL, Docker build, browser, and HCI artifact checks before Render deploys.

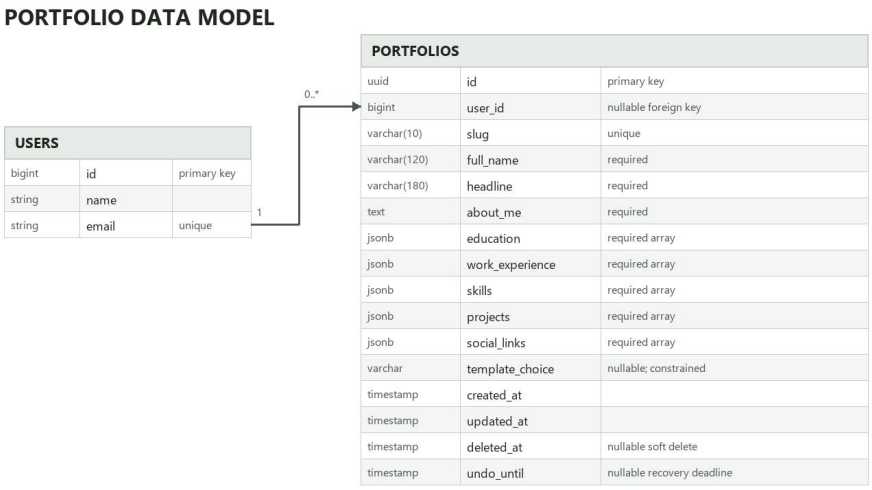
Table 3. Main workflow stages

Stage	User action	System response
1	Enter portfolio details	Form Request validates fields; the app stores a draft locally and shows inline feedback.
2	Compare and choose a template	The app previews the user's information and saves the selected supported template through a signed route.
3	Publish and share	Account owners find portfolios in My portfolios; guests receive a private signed management URL with a copy action.
4	Edit or change a template	An account owner uses the dashboard or a guest opens the private signed link to update the same portfolio record.
5	Delete or undo	The app soft-deletes the record, allows signed recovery for ten seconds, and sends a signed purge request when the undo period ends.

4. Database schema and data dictionary

4.1 Portfolio entity

Figure 2. Portfolio data model



Portfolio content is stored in `portfolios`; structured fields use JSONB in PostgreSQL and are cast to arrays by Eloquent. Nullable `user_id` links account-owned portfolios to `users`. Guest portfolios keep `user_id` null and remain manageable through their signed private link until claimed.

Table 4. `portfolios` data dictionary

Column	PostgreSQL type	Constraint	Description
id	uuid	Primary key	Application-generated portfolio identifier.
user_id	bigint	Nullable foreign key to users.id; null on user deletion	Account owner; null identifies a guest-owned portfolio.
slug	varchar(10)	Unique, required	Random ten-character route key used in public URLs.
full_name	varchar(120)	Required	Name displayed on the portfolio.
headline	varchar(180)	Required	Short professional tagline.
about_me	text	Required	Introduction, limited to 2,000 characters by request validation.
education	jsonb	Required array	Up to ten education entries with institution, qualification, and year.
work_experience	jsonb	Required array	Up to ten work entries with company, role, description, and month fields.
skills	jsonb	Required array, 1-30 entries	Skill names, up to 50 characters each.
projects	jsonb	Required array	Up to twenty entries with name, description, and optional HTTP/HTTPS URL.

Column	PostgreSQL type	Constraint	Description
social_links	jsonb	Required array	Up to twenty labeled HTTP/HTTPS links.
template_choice	varchar	Nullable; check constraint	Null before selection, otherwise Simple, Modern, or Creative.
created_at	timestamp	Nullable	Laravel creation timestamp.
updated_at	timestamp	Nullable	Laravel update timestamp.
deleted_at	timestamp	Nullable	Soft-delete marker; the record remains recoverable during the undo period.
undo_until	timestamp	Nullable	Deadline after which the scheduled purge may permanently delete the row.

PostgreSQL enforces the template choice with a CHECK constraint. SQLite migration tests use insert/update triggers to enforce the same allowed values. `Portfolio` has an explicit `$fillable` list; IDs, slugs, timestamps, and deletion fields cannot be mass assigned from request input. The factory creates test data and the idempotent seeder creates the fictional `portgen001` demo.

5. Key routes and controller logic

5.1 Route map

Table 5. Main HTTP routes

Method	URL	Purpose	Access
GET	/	Redirect to the portfolio form	Public
GET	/portfolios/create	Open the creation flow	Public
GET, POST	/login	Show the login form or authenticate a user	Guest only; POST rate limited
GET, POST	/register	Show the registration form or create an account	Guest only; POST rate limited
POST	/logout	End the current session	Authenticated
GET	/my-portfolios	List the authenticated user's portfolios	Authenticated
POST	/store	Validate and create the portfolio	Public, rate limited; authenticated creation assigns the owner
POST	/manage/{slug}/claim	Claim an unclaimed guest portfolio	Authenticated owner of signed guest link
GET	/portfolio/{slug}/select-template	Choose a presentation	Signed guest or account owner
POST	/portfolio/{slug}/save-template	Save an allowed template	Signed guest or account owner; rate limited
GET	/p/{slug}	Display the public portfolio	Public

Method	URL	Purpose	Access
GET	/manage/{slug}	Open owner controls	Signed guest or account owner
GET	/manage/{slug}/edit	Edit the portfolio	Signed guest or account owner
PUT	/manage/{slug}	Validate and update content	Signed guest or account owner; rate limited
DELETE	/manage/{slug}	Begin the delete and undo flow	Signed guest or account owner; rate limited
GET	/manage/{slug}/deleted	Display the deletion countdown	Signed guest or account owner
POST	/manage/{slug}/undo	Restore during the undo window	Signed guest or account owner; rate limited
DELETE	/manage/{slug}/purge	Purge after the undo period	Signed guest or account owner; rate limited
GET	/help	Read help and keyboard shortcuts	Public
GET	/up	Health check	Public health endpoint

Error preview routes exist only in local and test environments. The production routes return friendly 403, 404, 419, 429, and 500 views. State-changing routes retain CSRF protection and use Laravel throttling.

5.2 Controller and validation

`PortfolioController` implements create, store, template selection, template save, public show, manage, edit, update, delete, deleted-state display, undo, and purge. `StorePortfolioRequest` and `UpdatePortfolioRequest` share portfolio rules. `SaveTemplateRequest` restricts template values. URLs accept only HTTP and HTTPS schemes. User content is rendered through escaped Blade expressions.

Owner access uses either authenticated account ownership or a signed guest link. A valid guest owner can claim an unclaimed portfolio; claiming assigns its `user_id`, after which the old guest link no longer grants management access. Guest owners should retain their private link until they claim the portfolio. Portfolio ownership is assigned in server-side code rather than mass-assigned input.

Security middleware adds Content Security Policy, `X-Frame-Options`, `X-Content-Type-Options`, `Referrer-Policy`, and `Permissions-Policy`. `APP_DEBUG` is set to false in Render production configuration. Production also requires a real `APP_KEY` in the hosting secret store.

6. User interface design and HCI compliance

6.1 Interaction design

The creation flow uses three named stages and collapsible form sections. The browser stores recoverable draft fields in localStorage, excluding CSRF and method fields. A restore prompt appears on reload. Inline errors identify missing values; submit, copy, and deletion provide visible status. Users can operate the core flow with pointer controls or keyboard navigation and can open the shortcut dialog from the shared layout.

The three public templates group the same information differently. Simple uses a quiet neutral palette, Modern uses neighboring blue hues, and Creative uses warm cream with plum and magenta accents. All colors are defined as CSS custom properties in one token block with dark-mode overrides. The test computes contrast for configured text and interface pairs in both modes. It is a token-level regression test; it is not a manual audit of every rendered color combination.

6.2 Accessibility evidence

The Playwright suite tests the full keyboard create-and-publish journey, skip-link behavior, inline validation, local draft restoration, template changes, copy feedback, signed management, edit, delete and undo, help, and error screens. For each of 17 captured states it runs axe-core at 375 by 812 and 1280 by 800, checks visible control target dimensions, and checks the mobile layout at 320 pixels for horizontal overflow. The create screen is additionally inspected at 200% browser zoom. The automated suite reports no serious or critical axe violations. The suite does not claim complete WCAG conformance or replace assistive-technology user testing.

6.3 Screenshot evidence

Each of the 17 listed screen states has a mobile and desktop full-page image under `docs/screenshots/`. Captures use fictional profile content. The private signed owner URL is replaced in the saved screenshot with an invalid example value; the browser test still verifies that the actual link copies correctly before redaction. Each exported figure shows the 375 by 812 mobile viewport and 1280 by 800 desktop viewport crop, with the full-page captures retained in the repository. Each pair is annotated with its principal HCI evidence.

6.3.1 Creation and template selection

The first three figures show the form, template previews, and completion screen. They provide evidence for staged task structure, field guidance, visible status, and preview-based recognition.

Figure 3. Create form, step 1

Mobile: `docs/screenshots/mobile/step-1-your-info.png`

Desktop: `docs/screenshots/desktop/step-1-your-info.png`

HCI evidence: S1, S5, S8, N1, N2, N3, N4, H2, H5, H8, U1, U3, U6, U7, V1, V2, V3, V7, V8.

Figure 4. Template selection, step 2

Mobile: `docs/screenshots/mobile/step-2-choose-template.png`

Desktop: `docs/screenshots/desktop/step-2-choose-template.png`

HCI evidence: N2, N5, N3, H6, U3, C1, V3.

Figure 5. Completion and sharing, step 3

Mobile: `docs/screenshots/mobile/step-3-done-share.png`

Desktop: `docs/screenshots/desktop/step-3-done-share.png`

HCI evidence: S3, S4, H1, U4, C4, V7. The owner link is redacted in the saved image.

6.3.2 Public portfolio templates

Figure 6. Public Simple template

Mobile: `docs/screenshots/mobile/template-simple.png`

Desktop: `docs/screenshots/desktop/template-simple.png`

HCI evidence: S1, H4, H8, C1, C2, C3, V1, V2, V4, V5, V8.

Simple clarity-first design

Simple is the readability baseline for the student persona, including a portfolio with only a name, headline, About text, and one skill. The same hierarchy expands for a rich profile: the header and hero share one 1100px content edge; the hero pairs a short introduction with a portrait; optional sections and their navigation links appear only when data exists. The About section adds a facts list only when the portfolio supplies useful facts. Project cards use a three-column desktop grid and collapse for narrow screens.

The green palette and material colors are defined in `resources/css/tokens.css` for light, dark, and system modes. Simple uses monochromatic green, with raised and inset surfaces separated by tuned shadow pairs and 18px surface corners. Shadows decorate surfaces only. A one-pixel border, high-contrast text, solid primary action, visible selection check or underline, and token-based focus ring preserve meaning when shadows are removed. Increased-contrast and forced-colors modes use two-pixel borders without shadows. See the neumorphism limitation and mitigation.

Owners see the shared header's template chooser, Manage menu, and theme control; visitors see no owner actions and cannot use `?preview=` to change the saved template. Template choices preview without writing to the database until Save; Cancel returns to the saved style. The menu links to edit and manage pages, and gives a signed-link copy action only to guest owners. Account owners get My portfolios. These actions retain normal links and forms, and the header controls are hidden in print.

The synthetic minimal fixture uses a 240px patterned monogram instead of an empty portrait block. The rich fixture uses locally generated artwork, so the supplied design reference's photo and copy are not reused.

Figure 6a. Minimal Simple portfolio with owner controls

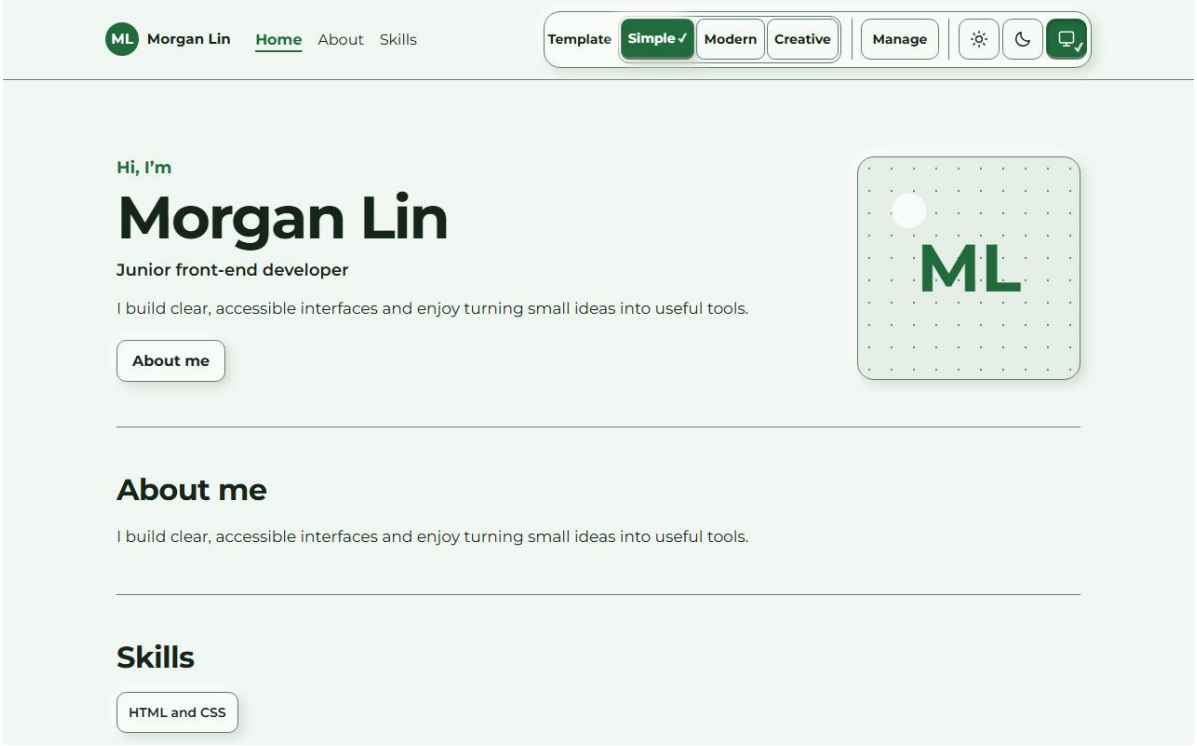


Figure 6b. Rich Simple portfolio with raised cards and inset active controls

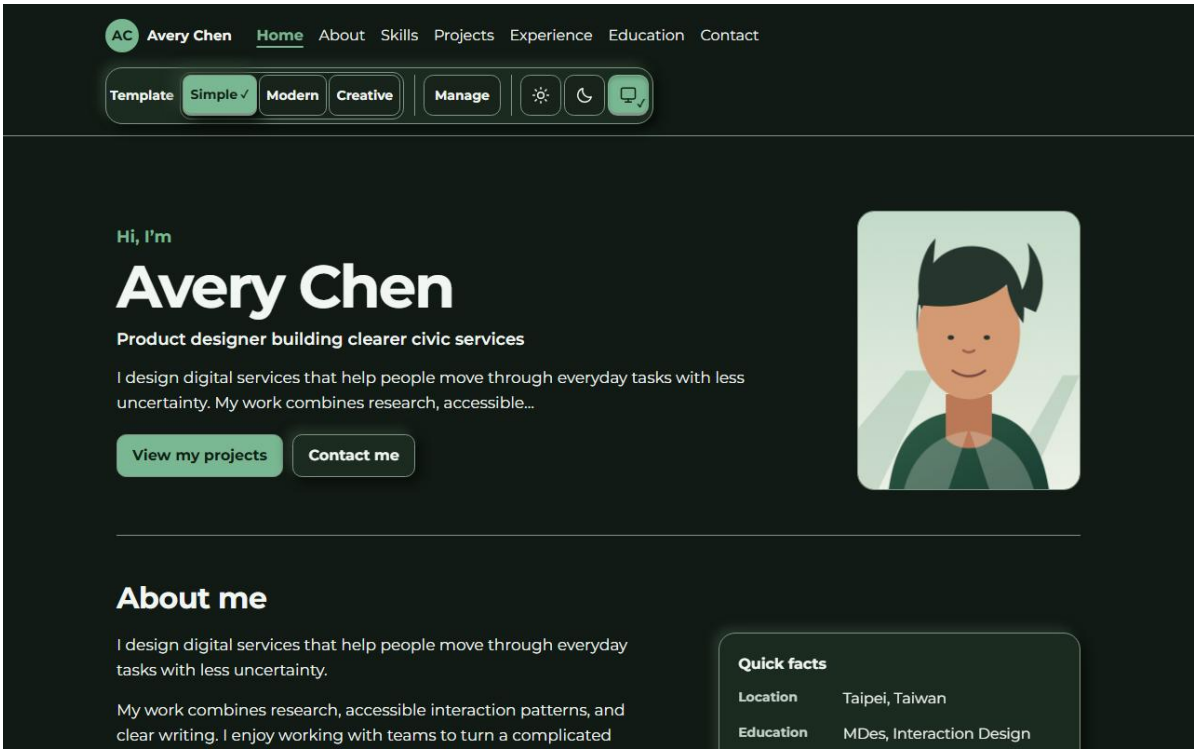
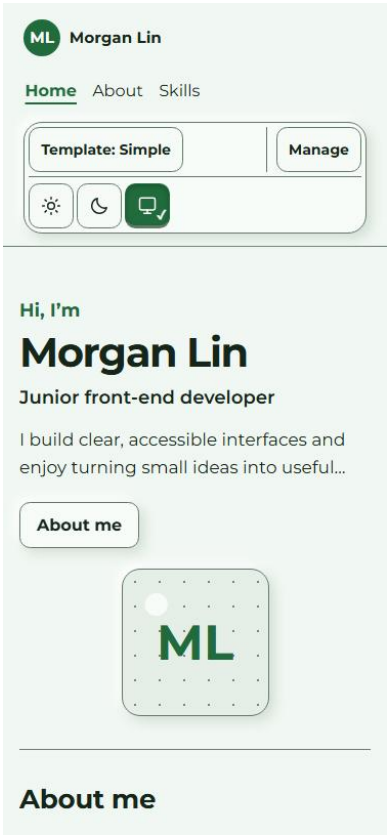


Figure 6c. Simple owner header and stacked layout at mobile width



The Playwright visual pass captured 60 combinations (three fixtures, five widths, two color modes, and owner/visitor views). All 60 axe runs reported zero serious or critical violations, and all 60 layouts passed the overflow check. The committed browser suite plus Simple owner and visual-review specs passed 6 tests; the live-deployment smoke was skipped locally. The focused Simple template spec checks keyboard selection, Escape, increased contrast, forced colors, reduced motion, and print visibility. `npm run build` passed, and `npm audit --omit=optional` reported zero vulnerabilities.

The full PHP suite and its `HCI_RELEASE=1` gate both exited successfully. The PHP 8.5 workstation reported 66 deprecation-marked cases and one ordinary pass, with 1,584 assertions in the regular run and 1,648 in the strict HCI run; these deprecations identify the existing MySQL SSL option constant and do not fail the suite. The strict tracker gate verified 64 of 64 HCI rows.

Figure 7. Public Modern template

Mobile: `docs/screenshots/mobile/template-modern.png`

Desktop: `docs/screenshots/desktop/template-modern.png`

HCI evidence: H6, C1, C2, C3, V3, V5, V6.

Figure 8. Public Creative template

Mobile: `docs/screenshots/mobile/template-creative.png`

Desktop: `docs/screenshots/desktop/template-creative.png`

HCI evidence: H6, C1, C2, C3, V4, V6, V8.

Figure 9. Creative template in dark mode

Mobile: `docs/screenshots/mobile/dark-mode-creative.png`

Desktop: `docs/screenshots/desktop/dark-mode-creative.png`

HCI evidence: C2, C3, V5. The CSS contrast test verifies configured palette pairs in light and dark mode.

6.3.3 Owner management and recovery

Figure 10. Private manage screen

Mobile: `docs/screenshots/mobile/manage.png`

Desktop: `docs/screenshots/desktop/manage.png`

HCI evidence: S1, S7, H1, H3, H4, N7, V1, V8.

Figure 11. Edit screen

Mobile: `docs/screenshots/mobile/edit.png`

Desktop: `docs/screenshots/desktop/edit.png`

HCI evidence: S1, N4, H3, U2, V2, V8.

Figure 12. Delete and undo

Mobile: `docs/screenshots/mobile/delete-undo.png`

Desktop: `docs/screenshots/desktop/delete-undo.png`

HCI evidence: S6, H3, H5, N6, U5. The undo window is ten seconds by requirement.

6.3.4 Help and errors

Figure 13. Help and FAQ

Mobile: `docs/screenshots/mobile/help-faq.png`

Desktop: `docs/screenshots/desktop/help-faq.png`

HCI evidence: N1, H2, H10, U1.

Figure 14. Keyboard shortcuts dialog

Mobile: `docs/screenshots/mobile/keyboard-shortcuts.png`

Desktop: `docs/screenshots/desktop/keyboard-shortcuts.png`

HCI evidence: S2, S3, H7, H10, U2, U6.

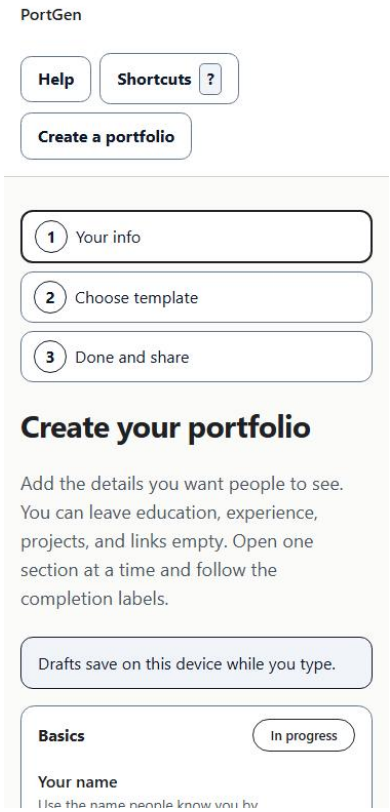
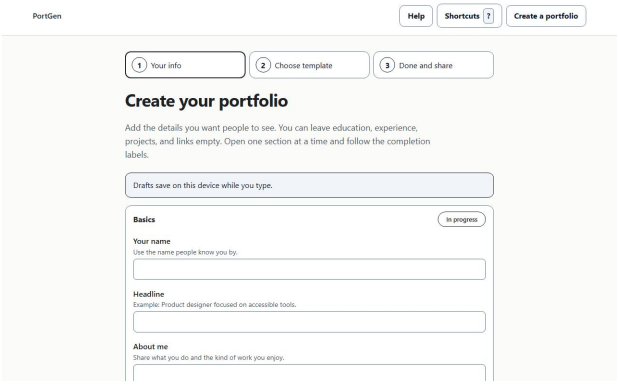
Figures 15-19. Friendly error states

For each error, the mobile and desktop captures appear in the exported paper: 403 `docs/screenshots/{mobile,desktop}/error-403.png`; 404 `error-404.png`; 419 `error-419.png`; 429 `error-429.png`; 500 `error-500.png`. The pages provide plain-language feedback and a return path. HCI evidence: S3, S5, N6, H1, H9, U4, C4.

6.3.5 Complete screenshot gallery

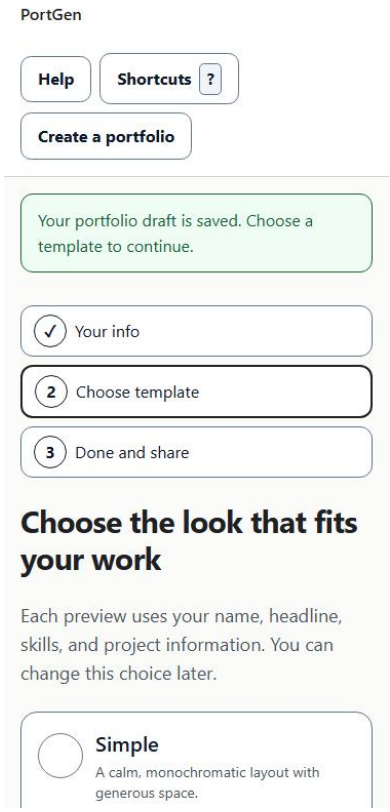
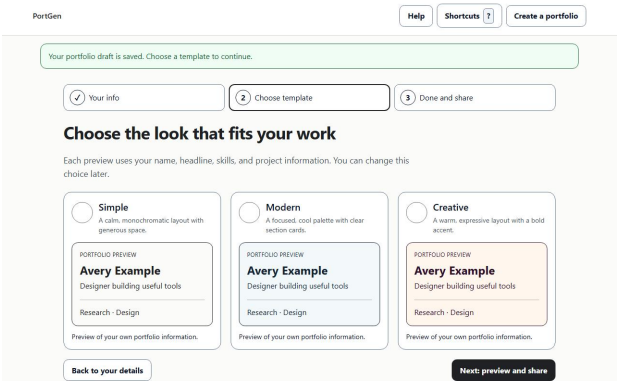
The following paired figures embed every mobile and desktop screen capture. Labels name the state shown and the primary principles illustrated.

Figure 20. Create form, step 1

Mobile view	Desktop view
Mobile - 375 x 812 	Desktop - 1280 x 800 

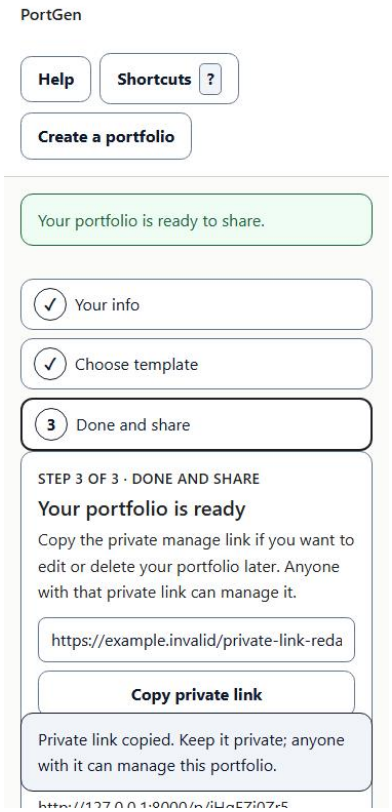
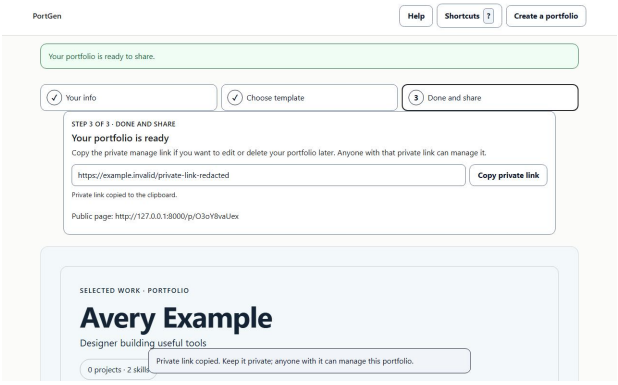
HCI principles: S8 Reduce short-term memory load; N4 Get the mappings right; H5 Error prevention; U7 Size and space for approach and use; V3 Visual hierarchy.

Figure 21. Template selection, step 2

Mobile view	Desktop view
Mobile - 375 x 812 	Desktop - 1280 x 800 

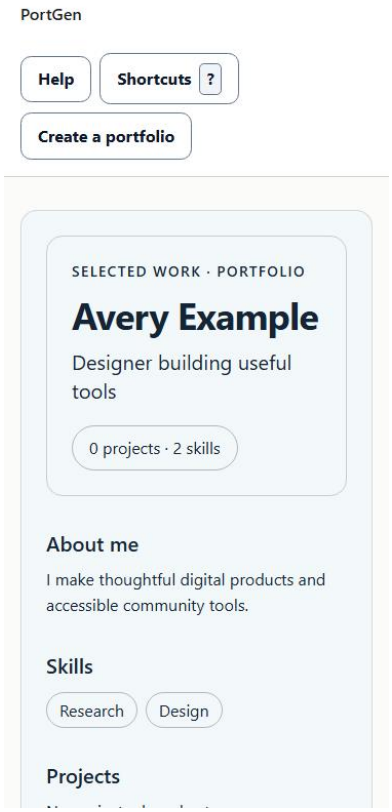
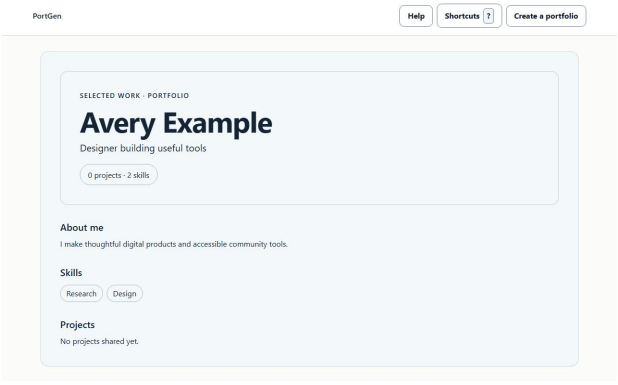
HCI principles: N2 Simplify task structure; H6 Recognition rather than recall; C1 Deliberate color scheme per template; V3 Visual hierarchy.

Figure 22. Completion and sharing, step 3

Mobile view	Desktop view
Mobile - 375 x 812 	Desktop - 1280 x 800 

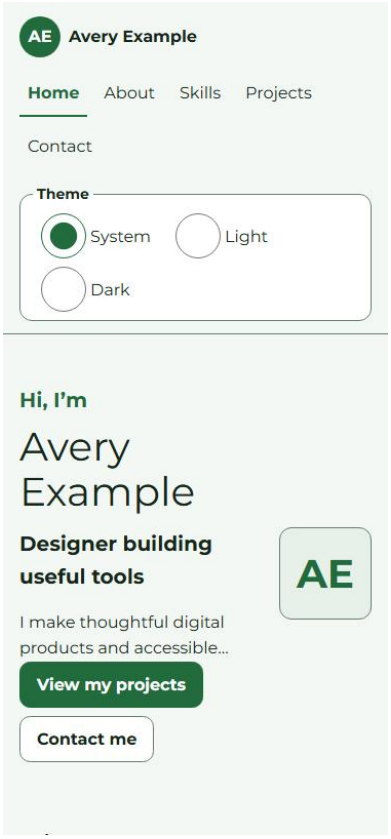
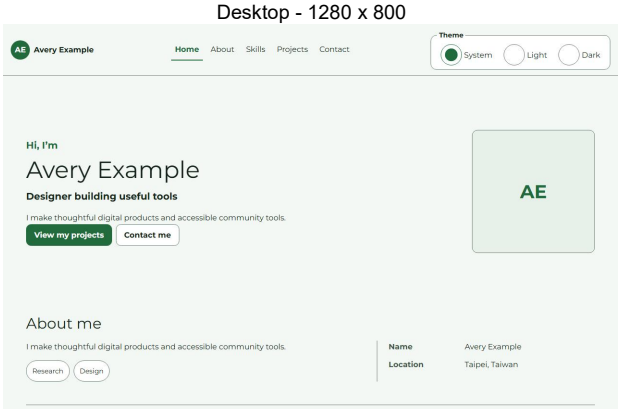
HCI principles: S3 Informative feedback; H1 Visibility of system status; C4 Never rely on color alone; V7 Dominance. the signed owner link is redacted in saved evidence

Figure 23. Modern portfolio preview

Mobile view	Desktop view
Mobile - 375 x 812 	Desktop - 1280 x 800 

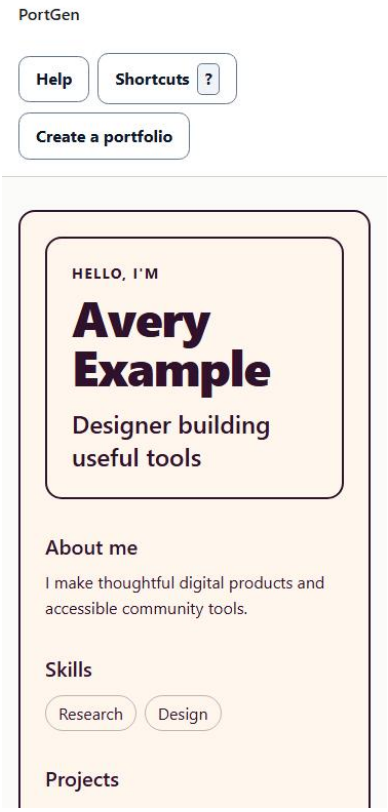
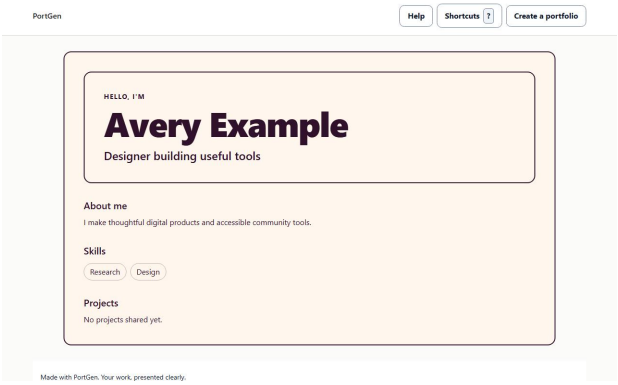
HCI principles: H6 Recognition rather than recall; C1 Deliberate color scheme per template; C3 Sufficient value contrast, WCAG AA; V5 Contrast.

Figure 24. Public Simple portfolio

Mobile view	Desktop view
Mobile - 375 x 812 	Desktop - 1280 x 800 

HCI principles: H8 Aesthetic and minimalist design; C1 Deliberate color scheme per template; V2 Gestalt principles; V4 Balance.

Figure 25. Public Creative portfolio

Mobile view	Desktop view
Mobile - 375 x 812 	Desktop - 1280 x 800 

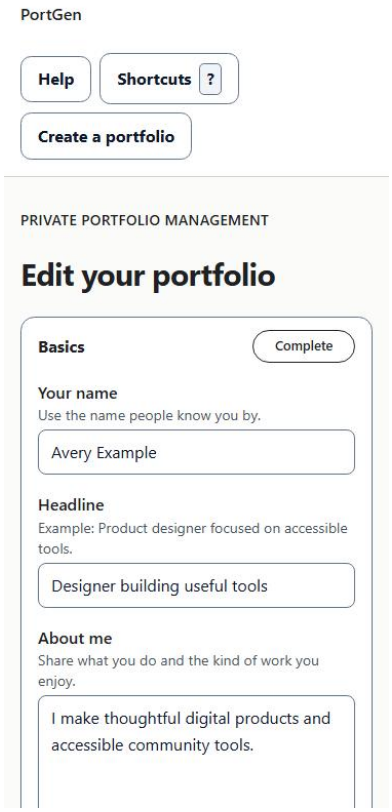
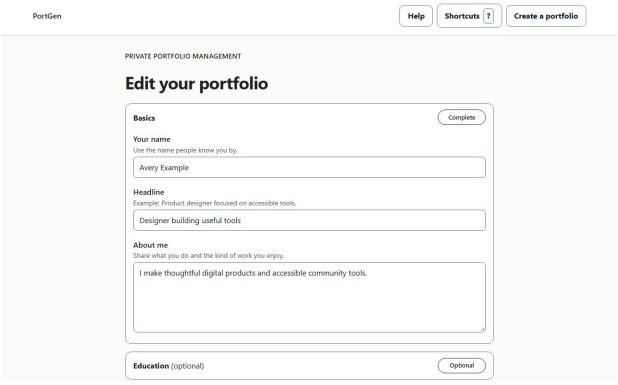
HCI principles: H6 Recognition rather than recall; C1 Deliberate color scheme per template; C2 Color temperature matches the template's message; V6 Scale.

Figure 26. Private portfolio management

Mobile view	Desktop view
Mobile - 375 x 812 PortGen Help Shortcuts ? Create a portfolio PRIVATE PORTFOLIO MANAGEMENT Manage portfolio for Avery Example This private link lets you edit or delete this portfolio. Save it somewhere private; anyone with the link can manage it. View the public portfolio Edit portfolio Change template Delete portfolio Made with PortGen. Your work, presented clearly.	Desktop - 1280 x 800 PortGen Help Shortcuts ? Create a portfolio PRIVATE PORTFOLIO MANAGEMENT Manage portfolio for Avery Example This private link lets you edit or delete this portfolio. Save it somewhere private; anyone with the link can manage it. View the public portfolio Edit portfolio Change template Delete portfolio Made with PortGen. Your work, presented clearly.

HCI principles: S7 Internal locus of control; H1 Visibility of system status; H3 User control and freedom; H4 Consistency and standards.

Figure 27. Edit portfolio

Mobile view	Desktop view
Mobile - 375 x 812 	Desktop - 1280 x 800 

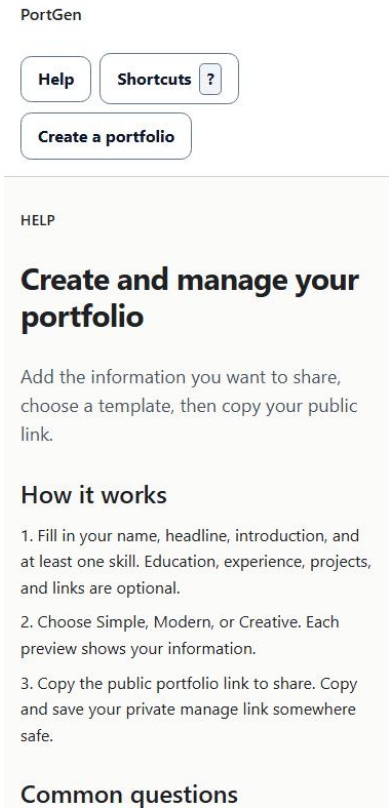
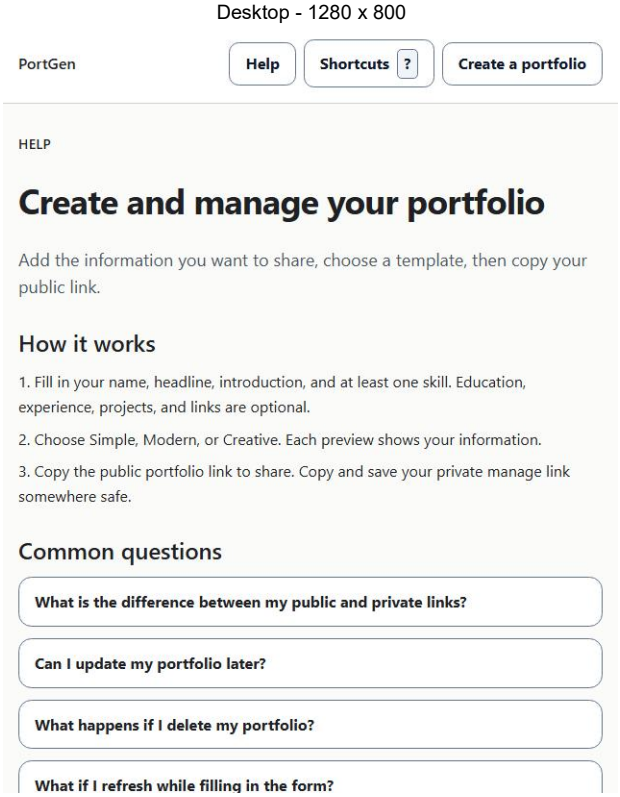
HCI principles: H3 User control and freedom; N4 Get the mappings right; U2 Flexibility in use; V2 Gestalt principles.

Figure 28. Delete and undo

Mobile view	Desktop view
Mobile - 375 x 812 PortGen Help Short PORTFOLIO DELETION Avery Example's portfol You have 9.704026 seconds to undo. After that, permanently removed. Undo deletion Create a different portfolio Made with PortGen. Your work, presented clearly.	Desktop - 1280 x 800 PortGen Help Shortcuts ? Create a portfolio PORTFOLIO DELETION Avery Example's portfolio is deleted You have 9.450185 seconds to undo. After that, the portfolio will be permanently removed. Undo deletion Create a different portfolio Made with PortGen. Your work, presented clearly.

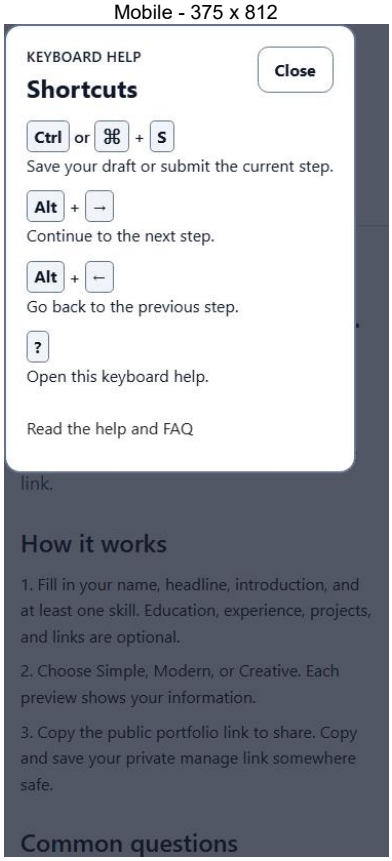
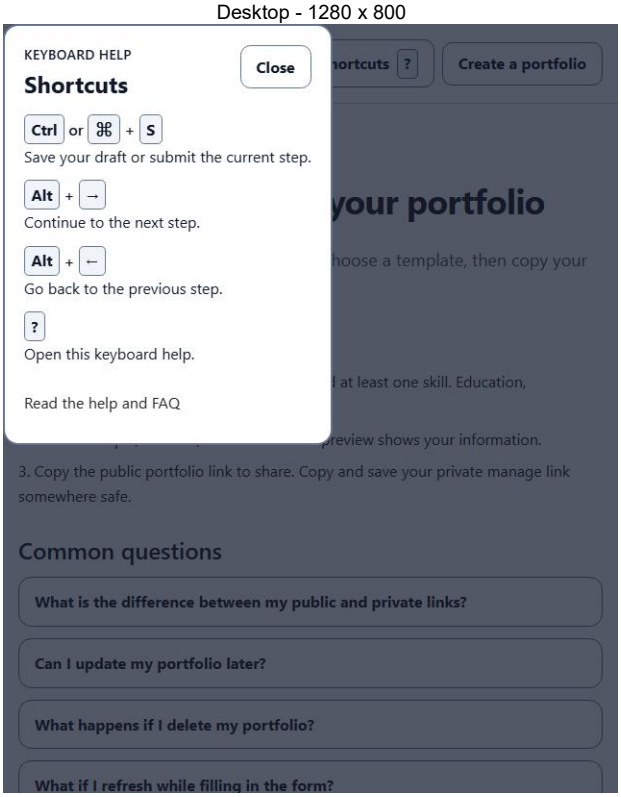
HCI principles: S6 Easy reversal of actions; H3 User control and freedom; U5 Tolerance for error. ten-second undo state

Figure 29. Help and FAQ

Mobile view	Desktop view
Mobile - 375 x 812 	Desktop - 1280 x 800 

HCI principles: N1 Knowledge in the world and in the head; H10 Help and documentation; U1 Equitable use.

Figure 30. Keyboard shortcuts dialog

Mobile view	Desktop view
Mobile - 375 x 812 	Desktop - 1280 x 800 

HCI principles: S2 Shortcuts for frequent users; H7 Flexibility and efficiency of use; U2 Flexibility in use.

Figure 31. Invalid private link, 403

Mobile view	Desktop view
Mobile - 375 x 812  	Desktop - 1280 x 800 

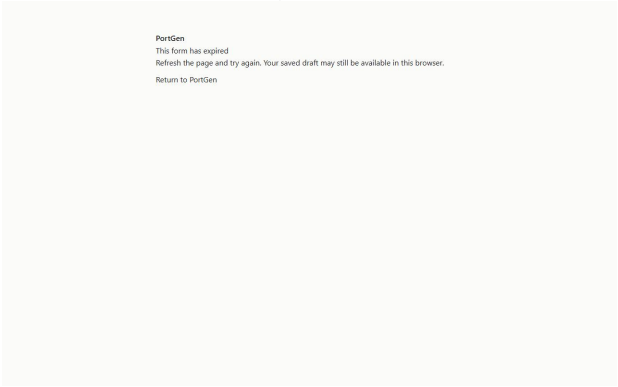
HCI principles: S3 Informative feedback; H9 Help users recognize, diagnose, recover from errors; C4 Never rely on color alone.

Figure 32. Missing portfolio, 404

Mobile view	Desktop view
Mobile - 375 x 812 	Desktop - 1280 x 800 

HCI principles: S3 Informative feedback; H9 Help users recognize, diagnose, recover from errors; C4 Never rely on color alone.

Figure 33. Expired form, 419

Mobile view	Desktop view
Mobile - 375 x 812 	Desktop - 1280 x 800 

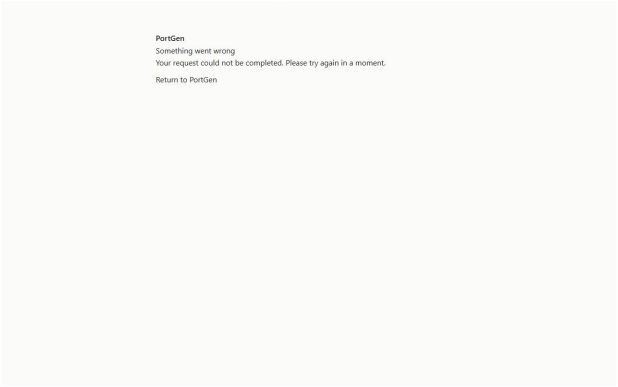
HCI principles: S3 Informative feedback; N6 Design for error; H9 Help users recognize, diagnose, recover from errors.

Figure 34. Rate limited, 429

Mobile view	Desktop view
Mobile - 375 x 812  	Desktop - 1280 x 800 

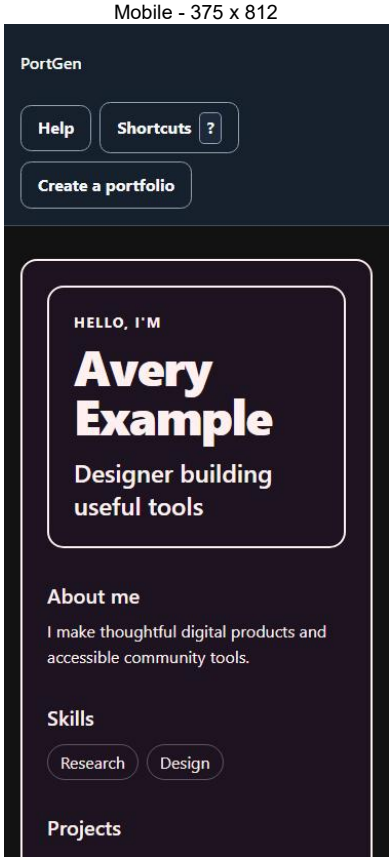
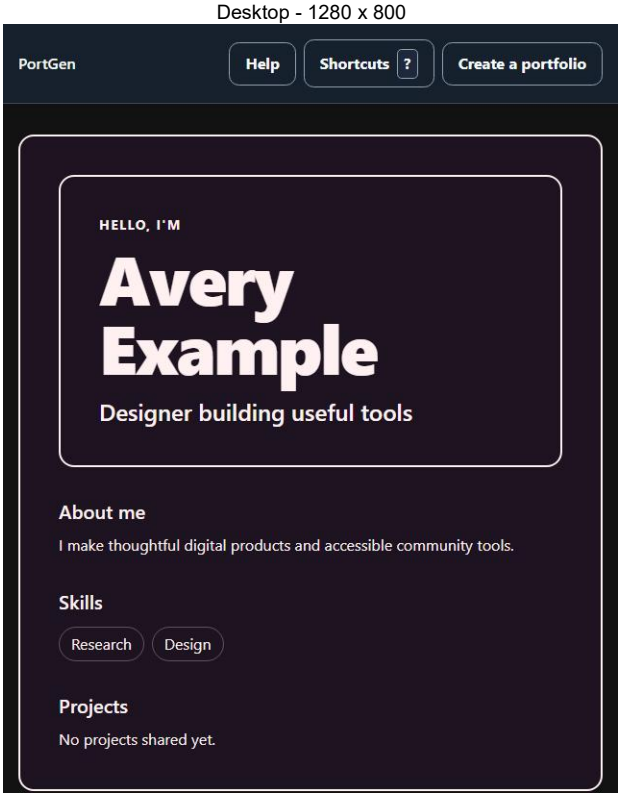
HCI principles: S3 Informative feedback; N6 Design for error; H9 Help users recognize, diagnose, recover from errors.

Figure 35. Server error, 500

Mobile view	Desktop view
Mobile - 375 x 812 	Desktop - 1280 x 800 

HCI principles: S3 Informative feedback; N6 Design for error; H9 Help users recognize, diagnose, recover from errors.

Figure 36. Creative portfolio in dark mode

Mobile view	Desktop view
Mobile - 375 x 812 	Desktop - 1280 x 800 

HCI principles: C2 Color temperature matches the template's message; C3 Sufficient value contrast, WCAG AA; V5 Contrast.

6.4 Full HCI tracker

The tables below reproduce all 48 rows from `docs/HCI_TRACKER.md`. All 48 rows now have a named test or committed screenshot, and the `HCI_RELEASE=1` release test passes. `Verified` means the row cites implementation evidence; it does not mean every criterion has been tested with participants.

Shneiderman's 8 Golden Rules

Table 6. Shneiderman's 8 Golden Rules HCI tracker

ID	Principle	Where applied	How applied	Evidence	Status
S1	Consistency	resources/views/layouts/app.blade.php; resources/css/app.css	Shared layout, navigation, spacing, and repeated button styles.	docs/screenshots/mobile/step-1-your-info.png; docs/screenshots/desktop/step-1-your-info.png	Verified
S2	Shortcuts for frequent users	resources/js/app.js; resources/views/layouts/app.blade.php	Discoverable save and step navigation shortcuts.	docs/screenshots/mobile/keyboard-shortcuts.png; docs/screenshots/desktop/keyboard-shortcuts.png; tests/e2e/portfolio-accessibility.spec.js::keyboard-first create, publish, manage, template, and delete-undo flows stay accessible	Verified
S3	Informative feedback	resources/views/layouts/app.blade.php; resources/views/portfolios/show.blade.php	Announce saved, published, copied-link, and error status in polite live regions.	docs/screenshots/mobile/step-3-done-share.png; docs/screenshots/desktop/step-3-done-share.png; PortfolioRoutesTest::test_template_selection_uses_and_preserves_the_private_link_for_the_creator	Verified
S4	Dialogs that yield closure	resources/views/portfolios/manage.blade.php; resources/views/portfolios/show.blade.php	Name the portfolio in the delete dialog and show the private owner link when creation finishes.	docs/screenshots/mobile/step-3-done-share.png; docs/screenshots/desktop/step-3-done-share.png	Verified
S5	Error prevention and simple error handling	app/Http/Requests/Concerns/HasPortfolioRules.php; resources/views/portfolios/form.blade.php; resources/js/app.js	Validate structured fields and safe URLs, then show specific inline recovery messages.	PortfolioRoutesTest::test_store_requires_a_skill_and_rejects_unsafe_urls; docs/screenshots/mobile/step-1-your-info.png	Verified
S6	Easy reversal of actions	routes/web.php; resources/views/portfolios/deleted.blade.php; app/Http/Controllers/PortfolioController.php	Offer a signed 10-second undo before permanent removal.	PortfolioRoutesTest::test_delete_can_be_undone_during_the_ten_second_window; docs/screenshots/mobile/delete-undo.png; docs/screenshots/desktop/delete-undo.png	Verified
S7	Internal locus of control	routes/web.php; app/Http/Controllers/PortfolioController.php	Require a private signed URL for management and keep transitions user initiated.	PortfolioRoutesTest::test_signed_manage_link_opens_and_unsigned_link_is_forbidden; docs/screenshots/mobile/manage.png	Verified
S8	Reduce short-term memory load	resources/views/portfolios/form.blade.php; resources/js/app.js	Group form fields, show completion labels, and offer local draft restoration.	docs/screenshots/mobile/step-1-your-info.png; tests/e2e/portfolio-accessibility.spec.js::keyboard-first create, publish, manage, template, and delete-undo flows stay accessible	Verified

Norman's 7 Principles

Table 7. Norman's 7 Principles HCI tracker

ID	Principle	Where applied	How applied	Evidence	Status
N1	Knowledge in the world and in the head	resources/views/portfolios/form.blade.php; resources/views/help.blade.php	Use examples and plain-language help beside unfamiliar fields.	docs/screenshots/mobile/help-faq.png; docs/screenshots/mobile/step-1-your-info.png	Verified
N2	Simplify task structure	resources/views/portfolios/stepper.blade.php; resources/views/portfolios/create.blade.php; resources/views/portfolios/select-template.blade.php	Divide creation into a visible three-step flow.	docs/screenshots/mobile/step-2-choose-template.png; docs/screenshots/desktop/step-2-choose-template.png	Verified
N3	Make things visible	resources/views/portfolios/stepper.blade.php; resources/js/app.js; resources/views/layouts/app.blade.php	Show current step, section completion, save status, and submit loading state.	docs/screenshots/mobile/step-2-choose-template.png; docs/screenshots/mobile/step-3-done-share.png	Verified
N4	Get the mappings right	resources/views/portfolios/form.blade.php; resources/views/portfolios/templates/sections.blade.php	Use familiar field names and group entered information under matching headings.	docs/screenshots/mobile/step-1-your-info.png	Verified
N5	Use constraints	app/Http/Requests/SaveTemplateRequest.php; app/Http/Requests/Concerns/HasPortfolioRules.php	Restrict template choice and URL schemes before storage.	PortfolioRoutesTest::test_template_choice_is_limited_to_the_three_supported_values; docs/screenshots/mobile/step-2-choose-template.png	Verified
N6	Design for error	resources/js/app.js; resources/views/portfolios/deleted.blade.php	Autosave drafts, explain validation failures, and provide delete recovery.	tests/e2e/portfolio-accessibility.spec.js::keyboard-first create, publish, manage, template, and delete-undo flows stay accessible; PortfolioRoutesTest::test_delete_can_be_undone_during_the_ten_second_window	Verified
N7	When all else fails, standardize	resources/views/layouts/app.blade.php; resources/css/app.css	Reuse semantic controls, navigation, and status components.	docs/screenshots/mobile/manage.png; docs/screenshots/mobile/edit.png	Verified

Nielsen's 10 Usability Heuristics

Table 8. Nielsen's 10 Usability Heuristics HCI tracker

ID	Principle	Where applied	How applied	Evidence	Status
H1	Visibility of system status	resources/views/layouts/app.blade.php; resources/views/portfolios/show.blade.php	Announce saved and published status and show the completion step.	docs/screenshots/mobile/step-3-done-share.png; PortfolioRoutesTest::test_template_selection_uses_and_preserves_the_private_link_for_the_creator	Verified
H2	Match between system and real world	resources/views/portfolios/form.blade.php; resources/views/help.blade.php	Use plain labels such as Skills, Projects, Save, and Next, with concrete examples.	docs/screenshots/mobile/help-faq.png; docs/screenshots/mobile/step-1-your-info.png	Verified
H3	User control and freedom	routes/web.php; resources/views/portfolios/deleted.blade.php; resources/js/app.js	Support back navigation, template changes, and deletion undo.	PortfolioRoutesTest::test_delete_can_be_undone_during_the_ten_second_window; docs/screenshots/mobile/delete-undo.png	Verified
H4	Consistency and standards	resources/views/layouts/app.blade.php; resources/css/app.css	Keep page structure, focus states, and control roles consistent.	docs/screenshots/mobile/manage.png; docs/screenshots/mobile/template-simple.png	Verified
H5	Error prevention	app/Http/Requests/Concerns/HasPortfolioRules.php; app/Http/Requests/SaveTemplateRequest.php	Validate required values, URL schemes, and supported templates before saving.	PortfolioRoutesTest::test_store_requires_a_skill_and_rejects_unsafe_urls; PortfolioRoutesTest::test_template_choice_is_limited_to_the_three_supported_values	Verified
H6	Recognition rather than recall	resources/views/portfolios/select-template.blade.php; resources/views/portfolios/form.blade.php	Preview each template with the person's own information and show field examples.	docs/screenshots/mobile/step-2-choose-template.png; docs/screenshots/mobile/template-modern.png	Verified
H7	Flexibility and efficiency of use	resources/js/app.js; resources/views/layouts/app.blade.php	Provide keyboard shortcuts alongside standard pointer controls.	docs/screenshots/mobile/keyboard-shortcuts.png; tests/e2e/portfolio-accessibility.spec.js::keyboard-first create, publish, manage, template, and delete-undo flows stay accessible	Verified
H8	Aesthetic and minimalist design	resources/views/portfolios/form.blade.php; resources/css/app.css	Show only the current creation task and group related information.	docs/screenshots/mobile/step-1-your-info.png; docs/screenshots/mobile/template-simple.png	Verified
H9	Help users recognize, diagnose, recover from errors	resources/views/errors/status.blade.php; resources/js/app.js; app/Http/Requests/Concerns/HasPortfolioRules.php	Show friendly server errors and specific inline field guidance.	docs/screenshots/mobile/error-403.png; docs/screenshots/mobile/error-500.png; PortfolioRoutesTest::test_store_requires_a_skill_and_rejects_unsafe_urls	Verified
H10	Help and documentation	resources/views/help.blade.php; resources/views/layouts/app.blade.php	Link a help page and shortcut dialog from every screen.	docs/screenshots/mobile/help-faq.png; docs/screenshots/mobile/keyboard-shortcuts.png	Verified

Universal Design

Table 9. Universal Design HCI tracker

ID	Principle	Where applied	How applied	Evidence	Status
U1	Equitable use	resources/views/layouts/app.blade.php; resources/css/app.css	Provide keyboard and screen-reader access to core tasks.	tests/e2e/portfolio-accessibility.spec.js::keyboard-first create, publish, manage, template, and delete-undo flows stay accessible; docs/screenshots/mobile/keyboard-shortcuts.png	Verified
U2	Flexibility in use	resources/js/app.js; resources/views/layouts/app.blade.php	Offer shortcuts and ordinary pointer controls.	docs/screenshots/mobile/keyboard-shortcuts.png; docs/screenshots/desktop/keyboard-shortcuts.png	Verified
U3	Simple and intuitive use	resources/views/portfolios/create.blade.php; resources/views/portfolios/select-template.blade.php; resources/views/portfolios/show.blade.php	Label the path from portfolio information through template choice to sharing.	docs/screenshots/mobile/step-1-your-info.png; docs/screenshots/mobile/step-2-choose-template.png	Verified
U4	Perceptible information	resources/css/app.css; resources/views/portfolios/form.blade.php	Use contrast-tested text and pair status colors with readable labels.	ColorContrastTest::test_light_and_dark_design_tokens_meet_wcag_contrast_thresholds; docs/screenshots/mobile/step-3-done-share.png	Verified
U5	Tolerance for error	routes/web.php; resources/views/portfolios/deleted.blade.php	Keep deleted content recoverable for 10 seconds, then permanently remove it.	PortfolioRoutesTest::test_delete_can_be_undone_during_the_ten_second_window; PortfolioModelTest::test_purge_command_removes_portfolios_after_the_undo_period; docs/screenshots/mobile/delete-undo.png	Verified
U6	Low physical effort	resources/css/app.css; resources/js/app.js	Support keyboard navigation and give controls generous spacing.	tests/e2e/portfolio-accessibility.spec.js::keyboard-first create, publish, manage, template, and delete-undo flows stay accessible; docs/screenshots/mobile/keyboard-shortcuts.png	Verified
U7	Size and space for approach and use	resources/css/app.css	Give interactive controls at least 44 CSS pixels and adapt layouts for narrow screens.	tests/e2e/portfolio-accessibility.spec.js::keyboard-first create, publish, manage, template, and delete-undo flows stay accessible; docs/screenshots/mobile/step-1-your-info.png	Verified

Color theory

Table 10. Color theory HCI tracker

ID	Principle	Where applied	How applied	Evidence	Status
C1	Deliberate color scheme per template	resources/css/app.css; docs/DECISIONS.md	Use monochromatic Simple neutrals, analogous blue Modern hues, and complementary plum/magenta with warm cream for Creative.	docs/screenshots/mobile/template-simple.png; docs/screenshots/mobile/template-modern.png; docs/screenshots/mobile/template-creative.png	Verified
C2	Color temperature matches the template's message	resources/css/app.css; docs/DECISIONS.md	Keep Simple restrained, Modern cool and calm, and Creative warm and expressive.	docs/screenshots/mobile/template-simple.png; docs/screenshots/mobile/template-modern.png; docs/screenshots/mobile/template-creative.png	Verified
C3	Sufficient value contrast, WCAG AA	resources/css/app.css; tests/Unit/ColorContrastTest.php	Check text pairs at 4.5:1 and control/focus boundaries at 3:1 in light and dark palettes.	ColorContrastTest::test_light_and_dark_design_tokens_meet_wcag_contrast_thresholds; docs/screenshots/mobile/dark-mode-creative.png	Verified
C4	Never rely on color alone	resources/views/portfolios/form.blade.php; resources/views/portfolios/deleted.blade.php; resources/css/app.css	Pair completion, error, success, and undo states with visible text.	PortfolioRoutesTest::test_store_requires_a_skill_and_rejects_unsafe_urls; docs/screenshots/mobile/error-429.png; docs/screenshots/mobile/step-3-done-share.png	Verified
C5	Define all colors as design tokens in one place	resources/css/app.css	Define app and template palette values in one root token block with dark overrides.	ColorContrastTest::test_light_and_dark_design_tokens_meet_wcag_contrast_thresholds; resources/css/app.css	Verified

Visual design principles

Table 11. Visual design principles HCI tracker

ID	Principle	Where applied	How applied	Evidence	Status
V1	Unity	resources/css/app.css; resources/views/layouts/app.blade.php	Reuse typography, spacing, and component styles throughout the app.	docs/screenshots/mobile/manage.png; docs/screenshots/mobile/template-simple.png	Verified
V2	Gestalt principles	resources/css/app.css; resources/views/portfolios/form.blade.php	Group related fields, repeat section styling, and align content.	docs/screenshots/mobile/step-1-your-info.png; docs/screenshots/mobile/template-creative.png	Verified
V3	Visual hierarchy	resources/css/app.css; resources/views/portfolios/stepper.blade.php	Use heading scale, step state, and spacing to guide the current task.	docs/screenshots/mobile/step-2-choose-template.png	Verified
V4	Balance	resources/css/app.css; resources/views/portfolios/templates/simple.blade.php	Balance primary content, supporting hints, and whitespace at each viewport.	docs/screenshots/mobile/template-simple.png; docs/screenshots/mobile/template-creative.png	Verified
V5	Contrast	resources/css/app.css; tests/Unit/ColorContrastTest.php	Separate text, controls, and focus states with measured contrast.	ColorContrastTest::test_light_and_dark_design_tokens_meet_wcag_contrast_thresholds; docs/screenshots/mobile/dark-mode-creative.png	Verified
V6	Scale	resources/css/app.css; resources/views/portfolios/templates/creative.blade.php	Use a consistent readable type and component scale with responsive headings.	docs/screenshots/mobile/template-creative.png; docs/screenshots/desktop/template-creative.png	Verified
V7	Dominance	resources/css/app.css; resources/views/portfolios/create.blade.php	Make the current heading and primary action the clearest elements.	docs/screenshots/mobile/step-3-done-share.png; docs/screenshots/mobile/step-1-your-info.png	Verified
V8	Similarity	resources/css/app.css; resources/views/layouts/app.blade.php	Style controls with the same role consistently throughout the application.	docs/screenshots/mobile/manage.png; docs/screenshots/mobile/edit.png	Verified

User needs

Table 12. User needs HCI tracker

ID	Principle	Where applied	How applied	Evidence	Status
R1	Documented personas and tasks	docs/USERS_AND_NEEDS.md	Describe student, freelancer, and job seeker contexts and tasks.	HciTrackerTest::test_documented_personas_and_core_tasks_cover_target_users; docs/USERS_AND_NEEDS.md	Verified
R2	Documented assumptions to validate	docs/USERS_AND_NEEDS.md	Separate assumptions from measured user research.	HciTrackerTest::test_user_research_assumptions_are_labeled_as_unvalidated; docs/USERS_AND_NEEDS.md	Verified
R3	Iteration log of changes from evaluation	docs/USERS_AND_NEEDS.md	Record evaluation evidence, findings, design changes, and follow-up work.	HciTrackerTest::test_iteration_log_records_evaluation_findings_and_design_changes; docs/USERS_AND_NEEDS.md; docs/COGNITIVE_WALKTHROUGH.md	Verified

7. Evaluation

7.1 Heuristic review

The single-reviewer expert inspection covers the 17 captured states against all ten Nielsen heuristics at mobile and desktop sizes. The screen-by-heuristic matrix, severity scale, findings, re-evaluation, and limitations are recorded in `docs/HEURISTIC_EVALUATION.md`. One minor finding remains: ten seconds is a short recovery period. A second cleanup finding was corrected by adding server-side scheduled purge and treating an already-purged record as completed. No severity 3 or 4 finding remains.

Table 13. Expert review findings and disposition

Finding	Severity	Disposition
The ten-second delete undo can be missed by a user who needs more time.	2	Retained to match the product requirement; ask classmates whether it is long enough.
Closing the delete screen could leave cleanup dependent on a later browser request.	2 before fix	Resolved with ten-second server-side purge scheduling and already-purged response handling; covered by purge command and schedule tests.

7.2 Cognitive walkthrough

This walkthrough applies the student, freelancer, and job-seeker personas to the implemented task path. It is an expert review based on the screens and route behavior, not observed participant behavior. The step-by-step record is in `docs/COGNITIVE_WALKTHROUGH.md`.

Table 14. Expert cognitive walkthrough

Persona	Task path reviewed	Expected user questions	Inspection result and limitation
Student	Add name, headline, introduction, skills, and optional education or course projects; select a template; copy the public URL.	“What information is required?”, “Can I leave work experience empty?”, “How do I share?”	Field guidance, optional section labels, a required skill message, template previews, and a completion screen answer the main questions. A classmate should validate terminology and whether the required fields feel reasonable.
Freelancer	Add a specialty and projects, preview styles, publish, and later update a project.	“Will my work look professional?”, “Which link is public?”, “Can I edit after sharing?”	The three previews use the same profile data; public and private links have separate labels; the private link opens guest edit and template controls, while account owners use My portfolios.
Job seeker	Add experience and skills, use keyboard navigation, publish a focused page, and test deletion recovery.	“Where am I in the process?”, “Can I correct a mistake?”, “Can I undo a delete?”	The stepper, field validation, keyboard shortcuts, edit page, and ten-second undo provide visible controls. Ten seconds may be too brief for some users and is an explicit study question.

7.3 Usability test plan

The kit in `docs/usability-testing/` contains a five-task moderator script, an observation sheet, a ten-item SUS questionnaire, and a results table. Participant count, task results, quotes, and SUS values remain marked “to be completed with classmates.” No user-study score is reported in this paper.

7.4 Iteration record

The R3 log in `docs/USERS_AND_NEEDS.md` records the draft restoration and repeat-row fixes observed during browser testing, and the server-side purge issue found during code inspection. The log distinguishes automation evidence from future classmate evaluation.

8. Deployment and infrastructure

8.1 Release configuration

`Dockerfile` builds Composer dependencies, Vite assets, and a PHP 8.3 Apache image. `docker/start.sh` binds Apache to Render's `PORT`, applies additive migrations, seeds the fictional demo, and builds Laravel's configuration, route, and view caches. It also supervises the sub-minute scheduler. Render's free plan does not support a pre-deploy command, so migration runs at service startup. Free web services sleep after inactivity and have ephemeral local storage; Render Postgres retains application data, sessions, and cache until the Free database expires after 30 days. Render provides a 14-day grace period to upgrade an expired database before deleting it. `.github/workflows/ci.yml` audits PHP dependencies and builds the Render image; `.github/workflows/smoke.yml` checks the configured live URL after CI.

Figure 37. CI and deployment flow



Table 15. Deployment environment variables

Variable	Value source
APP_ENV, APP_DEBUG, APP_TIMEZONE	Render production environment; production, false, and Asia/Taipei.
APP_KEY	Render-generated 256-bit base64 secret, normalized to Laravel's base64: form when needed.
APP_URL	Render's assigned HTTPS URL, referenced from RENDER_EXTERNAL_URL.
DB_CONNECTION	pgsql.
DB_URL	Render Postgres private connection string supplied through the Blueprint.
SESSION_DRIVER, CACHE_STORE, QUEUE_CONNECTION	Database, database, and sync for the stateless container.
LOG_CHANNEL, SESSION_SECURE_COOKIE	stderr and true for runtime logs and HTTPS session cookies.
PORTGEN_LIVE_URL	GitHub repository variable for the smoke workflow, set only after a real deployment.

Render's Blueprint supplies the generated Postgres connection string without committing database credentials. Production migrations must be additive; `migrate:fresh` and `db:wipe` are reserved for the isolated CI database. The test harness must reject production database URLs. Render's free instances have ephemeral local storage, so the session and cache tables share state through PostgreSQL. The Free Postgres database is intended only for a demonstration because Render expires it 30 days after creation and deletes its data after a further 14-day upgrade grace period.

8.2 Deployment status and blockers

The source repository is public at `https://github.com/crazyfrogfrog84-dotcom/portgen`. The Render Blueprint defines service `portgen-m1uh` and database `portgen-m1uh-db`; the service is live and deploys from `main` at `https://portgen-m1uh-if1e.onrender.com`. Direct checks on 2026-10-20 returned HTTP 200 for `/up`, `/portfolios/create`, and `/p/portgen001`. The latest main CI run passed. Its following Live smoke run failed during health/page preflight before the create-and-render test; later direct checks passed.

Laravel 12.70.0 replaces the unsupported Laravel 11 dependency set and `composer audit --locked` reports no advisories. The Render service uses a generated `APP_KEY`, binds its database through the Blueprint, and deploys after CI checks pass. Production schema changes use additive migrations only. The Free database will require an upgrade or data export before its 30-day expiration and 14-day grace period end.

9. Testing and technical achievements

9.1 Local command results

Table 16. Verification executed in this workspace

Command	Result
php artisan test	Exit code 0; 1,584 assertions; 66 deprecation-marked cases and 1 ordinary pass under PHP 8.5.
./vendor/bin/pint --test	Passed.
npm run build	Passed with Vite 8.3.4.
npm audit --omit=optional	Zero vulnerabilities.
Checked-in browser suite plus Simple-focused specs	6 passed; 1 live-only smoke test skipped locally. The separate Simple visual run audited all 60 viewport/theme/owner combinations.
php artisan schedule:list	Shows portfolios:purge-expired every ten seconds.
HCI_RELEASE=1 php artisan test	Exit code 0; 1,648 assertions; all 64 HCI tracker rows verified.
composer audit --locked	Exit code 0; no security vulnerability advisories found after upgrading to Laravel 12.70.0.
php scripts/check-secrets.php --worktree	Passed.

The workstation uses PHP 8.5, not the required PHP 8.3 runtime, so PHPUnit reports deprecations from the existing MySQL SSL option constant. CI uses PHP 8.3 and PostgreSQL 16 and builds the Render image.

9.2 Database, browser, and security checks

Feature tests cover portfolio validation, session registration/login/logout, ownership and claiming, dashboard behavior, signed guest access, template controls, escaping, rate limits, demo seeding, deletion/undo, and purge behavior. Model tests cover UUID and slug behavior, mass-assignment protection, uniqueness, JSON arrays, template constraints, seeder idempotence, and scheduled cleanup. Browser checks cover keyboard flow, draft state, responsive widths, theme modes, error screens, and the Simple owner controls; CI runs the database gates on an isolated PostgreSQL 16 service.

The implementation includes CSRF, rate limits on state-changing routes, account ownership and signed guest URLs, strict URL schemes, escaped user content, friendly error views, security headers, an explicit mass-assignment allowlist, and a repository secret scanner. A guest signed URL is a bearer credential and should be kept private. The live smoke workflow checks the service endpoints and then creates and renders a throwaway portfolio after successful CI.

10. User guide

10.1 Create and publish

1. Open `/portfolios/create` and enter your name, headline, and introduction.
2. Expand the optional Education, Experience, Projects, and Links sections as needed. Add at least one skill.
3. Use Next to compare previews populated with your information.
4. Choose Simple, Modern, or Creative, then finish the sharing step.
5. Copy the public URL to share your portfolio. Save the separate private management link in a secure place.

10.2 Manage and edit

Open My portfolios to manage an account-owned portfolio. A guest can open the private signed link to view its controls, edit details, or preview a template. Anyone who has a guest link can manage that portfolio, so do not share it publicly.

10.3 Delete and undo

Choose Delete portfolio and review the named confirmation. The deletion screen provides a ten-second Undo control. After the window ends, the server scheduler permanently removes the record. The local copy in the browser is not a backup.

10.4 Keyboard shortcuts

Use Ctrl/Cmd+S to save a draft or submit the current step, Alt+Right/Left to move through steps, and `?` to open the shortcut dialog. The shared layout also links the Help and FAQ page.

11. Maintenance and future work

11.1 Run locally

1. Install PHP 8.3, Composer, Node.js 24, and npm.
2. Run `composer install` and `npm ci`.
3. Copy `.env.example` to `.env`, generate an application key with `php artisan key:generate`, and configure an isolated local database.
4. Run `php artisan migrate --seed` and `npm run build`.
5. Start the app with `php artisan serve` and the front-end watcher with `npm run dev`.
6. Run `php artisan test`, `./vendor/bin/pint --test`, `npm audit --omit=optional`, and `npm run test:e2e` for the local checks.

Never point test commands or `migrate:fresh` at a production database. For a supported public launch, upgrade the framework dependency to a currently supported Laravel release and re-run the full CI gate.

11.2 Future work

- Keep Laravel 12 dependencies patched and plan the Laravel 13 upgrade before Laravel 12's security-fix date.
- Keep the Render service `portgen-m1uh` healthy and keep the live create-and-render smoke checks green.
- Upgrade or export the Render Free Postgres database before its 30-day expiration and 14-day grace period end if any data must be retained.
- Conduct the five-task classmate study and add real observation and SUS results.
- Consider profile photos and project images only after launch readiness; validate type, size, alt text, preview, storage, and deletion flows.
- Evaluate skill-logo suggestions with an explicit text fallback and no dependency on color or imagery.

12. Appendices

Appendix A. Decisions

Table 17. Dated implementation decisions

Date	Decision	Reason
2026-10-19	Initially retain Laravel 11 to match the fixed stack in the brief; later superseded by the security upgrade decision below.	The framework's security-fix window had ended and Composer reported four advisories, so the initial decision was changed before deployment.
2026-10-19	Upgrade Laravel 11 to Laravel 12.70.0 and PHPUnit 11 before deployment.	Laravel 11's security-fix window ended on 2026-03-12 and the locked framework had four advisories. The official upgrade guide describes Laravel 12 as a low-impact update; the patched release removes the current Composer advisories while retaining PHP 8.3 compatibility.
2026-10-19	Set Composer's platform PHP version to 8.3.	The required CI and production runtime are PHP 8.3, while the workstation currently runs PHP 8.5. This keeps dependency resolution compatible with the requested runtime.
2026-10-19	Use Tailwind CSS 4 through its official Vite plugin and use Playwright with axe-core for browser checks.	These match the required build and browser-testing stack; the packages are dev dependencies and are needed for the requested UI and accessibility gates.
2026-10-19	Upgrade Vite and laravel-vite-plugin to versions compatible with Node 24 and Vite 8 after npm audit found vulnerable esbuild/Vite dependencies.	The existing Laravel 11 skeleton pinned Vite 5, which produced moderate and high npm audit findings. The current Laravel Vite plugin supports Vite 8; the updated dependency tree reports zero npm vulnerabilities.
2026-10-19	Use neutral monochrome tokens for Simple, cool neighboring blue hues for Modern, and warm cream with plum/magenta accents for Creative.	The palettes distinguish quiet, focused, and expressive presentation styles while keeping text and controls readable in both color schemes; ColorContrastTest measures the defined foreground/background pairs.
2026-10-19	Use a deterministic fictional public demo slug (portgen001) and an idempotent seeder for the live smoke check.	The deployment smoke workflow needs a stable public URL to verify that migrations, seeding, and public rendering work together without using real user data.
2026-10-19	Keep local error-preview routes available only in local and testing environments.	Playwright needs to inspect every friendly server-error screen, while production should expose only application routes.
2026-10-19	Run local Playwright with a dedicated temporary SQLite file and reserve PostgreSQL browser tests for the isolated CI service.	This machine has no Docker, PostgreSQL server, or GitHub runner access. The browser setup overrides DB connection settings and refuses a PostgreSQL target unless it is explicitly marked as the local portgen_test service.
2026-10-19	Use Render's Docker web service with Apache as the primary host.	The requested deployment target is Render; its Docker runtime supports the committed Apache image and uses the standard PORT variable. This database plan was superseded by the Render Free Postgres demonstration setup on 2026-10-20.

Date	Decision	Reason
2026-10-20	Use Render Free Postgres for the PortGen demonstration deployment.	The user requested a free deployment. Render Free Postgres keeps demo data separate from other projects, but expires after 30 days and deletes data after a 14-day grace period unless upgraded.
2026-10-19	Name the requested Render service portgen-m1uh and deploy the codex/render-deployment branch after its CI checks pass.	This matches the requested live URL slug and keeps the public service on a branch with CI checks before it is promoted to main.
2026-10-19	Keep Laravel's ten-second deletion undo and run its scheduler in the Render container.	The owner may close the browser during the undo window; Laravel's schedule:work performs cleanup while the service is running, and the signed browser purge provides request-driven cleanup.
2026-10-19	Use SQLite in memory only for the initial scaffold checks; use the dedicated local Postgres 16 service for application database tests and CI.	This avoids any accidental connection to production and allows the scaffold gate to run before a Docker service is available.
2026-10-19	Run the required Postgres 16 migration, rollback, and feature gates in the CI service; run only SQLite-backed migration/model tests on this workstation.	Docker, psql, and a local PostgreSQL service are absent, and the current Windows account cannot install a PostgreSQL service. The CI job declares its own ephemeral Postgres 16 service. Local SQLite results do not count as the Postgres gate.
2026-10-19	Run expired portfolio cleanup every ten seconds in the Render container.	The required ten-second undo period must also finish when an owner closes the browser. Laravel supports sub-minute schedules through schedule:work; the signed purge request also completes cleanup when the owner remains on the page.
2026-10-19	Keep the usability study kit open for classmates and report no participant scores until sessions have been run.	The implementation has expert inspection and automated checks, but no real participant data. Blank observation and results fields prevent fabricated findings.
2026-10-19	Store Mermaid source and a neutral theme configuration next to rendered diagrams and use the diagrams in the system paper.	Source files keep architecture, data, and release flow editable; rendered PNGs allow diagrams to be embedded in the DOCX and PDF. Mermaid CLI is used only as an authoring tool and is not added to application dependencies.

Appendix B. Glossary

Bearer link: A private URL whose possession grants access to a resource.

JSONB: PostgreSQL's binary JSON data type for structured values.

Signed URL: A URL with a cryptographic signature that detects tampering.

Soft delete: Marking a row as deleted while retaining it for a recovery period.

SUS: System Usability Scale, a standardized ten-item questionnaire.

WCAG: Web Content Accessibility Guidelines published by W3C.

Appendix C. References

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11. Render. “Deploy for Free.” Official documentation.

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Appendix D. HCI tracker copy

This copy matches the 48 implementation rows in section 6.4 and `docs/HCI_TRACKER.md`.

Shneiderman's 8 Golden Rules

Table 18. Shneiderman's 8 Golden Rules HCI tracker

ID	Principle	Where applied	How applied	Evidence	Status
S1	Consistency	resources/views/layouts/app.blade.php; resources/css/app.css	Shared layout, navigation, spacing, and repeated button styles.	docs/screenshots/mobile/step-1-your-info.png; docs/screenshots/desktop/step-1-your-info.png	Verified
S2	Shortcuts for frequent users	resources/js/app.js; resources/views/layouts/app.blade.php	Discoverable save and step navigation shortcuts.	docs/screenshots/mobile/keyboard-shortcuts.png; docs/screenshots/desktop/keyboard-shortcuts.png; tests/e2e/portfolio-accessibility.spec.js::keyboard-first create, publish, manage, template, and delete-undo flows stay accessible	Verified
S3	Informative feedback	resources/views/layouts/app.blade.php; resources/views/portfolios/show.blade.php	Announce saved, published, copied-link, and error status in polite live regions.	docs/screenshots/mobile/step-3-done-share.png; docs/screenshots/desktop/step-3-done-share.png; PortfolioRoutesTest::test_template_selection_uses_and_preserves_the_private_link_for_the_creator	Verified
S4	Dialogs that yield closure	resources/views/portfolios/manage.blade.php; resources/views/portfolios/show.blade.php	Name the portfolio in the delete dialog and show the private owner link when creation finishes.	docs/screenshots/mobile/step-3-done-share.png; docs/screenshots/desktop/step-3-done-share.png	Verified
S5	Error prevention and simple error handling	app/Http/Requests/Concerns/HasPortfolioRules.php; resources/views/portfolios/form.blade.php; resources/js/app.js	Validate structured fields and safe URLs, then show specific inline recovery messages.	PortfolioRoutesTest::test_store_requires_a_skill_and_rejects_unsafe_urls; docs/screenshots/mobile/step-1-your-info.png	Verified
S6	Easy reversal of actions	routes/web.php; resources/views/portfolios/deleted.blade.php; app/Http/Controllers/PortfolioController.php	Offer a signed 10-second undo before permanent removal.	PortfolioRoutesTest::test_delete_can_be_undone_during_the_ten_second_window; docs/screenshots/mobile/delete-undo.png; docs/screenshots/desktop/delete-undo.png	Verified
S7	Internal locus of control	routes/web.php; app/Http/Controllers/PortfolioController.php	Require a private signed URL for management and keep transitions user initiated.	PortfolioRoutesTest::test_signed_manage_link_opens_and_unsigned_link_is_forbidden; docs/screenshots/mobile/manage.png	Verified
S8	Reduce short-term memory load	resources/views/portfolios/form.blade.php; resources/js/app.js	Group form fields, show completion labels, and offer local draft restoration.	docs/screenshots/mobile/step-1-your-info.png; tests/e2e/portfolio-accessibility.spec.js::keyboard-first create, publish, manage, template, and delete-undo flows stay accessible	Verified

Norman's 7 Principles

Table 19. Norman's 7 Principles HCI tracker

ID	Principle	Where applied	How applied	Evidence	Status
N1	Knowledge in the world and in the head	resources/views/portfolios/form.blade.php; resources/views/help.blade.php	Use examples and plain-language help beside unfamiliar fields.	docs/screenshots/mobile/help-faq.png; docs/screenshots/mobile/step-1-your-info.png	Verified
N2	Simplify task structure	resources/views/portfolios/stepper.blade.php; resources/views/portfolios/create.blade.php; resources/views/portfolios/select-template.blade.php	Divide creation into a visible three-step flow.	docs/screenshots/mobile/step-2-choose-template.png; docs/screenshots/desktop/step-2-choose-template.png	Verified
N3	Make things visible	resources/views/portfolios/stepper.blade.php; resources/js/app.js; resources/views/layouts/app.blade.php	Show current step, section completion, save status, and submit loading state.	docs/screenshots/mobile/step-2-choose-template.png; docs/screenshots/mobile/step-3-done-share.png	Verified
N4	Get the mappings right	resources/views/portfolios/form.blade.php; resources/views/portfolios/templates/sections.blade.php	Use familiar field names and group entered information under matching headings.	docs/screenshots/mobile/step-1-your-info.png	Verified
N5	Use constraints	app/Http/Requests/SaveTemplateRequest.php; app/Http/Requests/Concerns/HasPortfolioRules.php	Restrict template choice and URL schemes before storage.	PortfolioRoutesTest::test_template_choice_is_limited_to_the_three_supported_values; docs/screenshots/mobile/step-2-choose-template.png	Verified
N6	Design for error	resources/js/app.js; resources/views/portfolios/deleted.blade.php	Autosave drafts, explain validation failures, and provide delete recovery.	tests/e2e/portfolio-accessibility.spec.js::keyboard-first create, publish, manage, template, and delete-undo flows stay accessible; PortfolioRoutesTest::test_delete_can_be_undone_during_the_ten_second_window	Verified
N7	When all else fails, standardize	resources/views/layouts/app.blade.php; resources/css/app.css	Reuse semantic controls, navigation, and status components.	docs/screenshots/mobile/manage.png; docs/screenshots/mobile/edit.png	Verified

Nielsen's 10 Usability Heuristics

Table 20. Nielsen's 10 Usability Heuristics HCI tracker

ID	Principle	Where applied	How applied	Evidence	Status
H1	Visibility of system status	resources/views/layouts/app.blade.php; resources/views/portfolios/show.blade.php	Announce saved and published status and show the completion step.	docs/screenshots/mobile/step-3-done-share.png; PortfolioRoutesTest::test_template_selection_uses_and_preserves_the_private_link_for_the_creator	Verified
H2	Match between system and real world	resources/views/portfolios/form.blade.php; resources/views/help.blade.php	Use plain labels such as Skills, Projects, Save, and Next, with concrete examples.	docs/screenshots/mobile/help-faq.png; docs/screenshots/mobile/step-1-your-info.png	Verified
H3	User control and freedom	routes/web.php; resources/views/portfolios/deleted.blade.php; resources/js/app.js	Support back navigation, template changes, and deletion undo.	PortfolioRoutesTest::test_delete_can_be_undone_during_the_ten_second_window; docs/screenshots/mobile/delete-undo.png	Verified
H4	Consistency and standards	resources/views/layouts/app.blade.php; resources/css/app.css	Keep page structure, focus states, and control roles consistent.	docs/screenshots/mobile/manage.png; docs/screenshots/mobile/template-simple.png	Verified
H5	Error prevention	app/Http/Requests/Concerns/HasPortfolioRules.php; app/Http/Requests/SaveTemplateRequest.php	Validate required values, URL schemes, and supported templates before saving.	PortfolioRoutesTest::test_store_requires_a_skill_and_rejects_unsafe_urls; PortfolioRoutesTest::test_template_choice_is_limited_to_the_three_supported_values	Verified
H6	Recognition rather than recall	resources/views/portfolios/select-template.blade.php; resources/views/portfolios/form.blade.php	Preview each template with the person's own information and show field examples.	docs/screenshots/mobile/step-2-choose-template.png; docs/screenshots/mobile/template-modern.png	Verified
H7	Flexibility and efficiency of use	resources/js/app.js; resources/views/layouts/app.blade.php	Provide keyboard shortcuts alongside standard pointer controls.	docs/screenshots/mobile/keyboard-shortcuts.png; tests/e2e/portfolio-accessibility.spec.js::keyboard-first create, publish, manage, template, and delete-undo flows stay accessible	Verified
H8	Aesthetic and minimalist design	resources/views/portfolios/form.blade.php; resources/css/app.css	Show only the current creation task and group related information.	docs/screenshots/mobile/step-1-your-info.png; docs/screenshots/mobile/template-simple.png	Verified
H9	Help users recognize, diagnose, recover from errors	resources/views/errors/status.blade.php; resources/js/app.js; app/Http/Requests/Concerns/HasPortfolioRules.php	Show friendly server errors and specific inline field guidance.	docs/screenshots/mobile/error-403.png; docs/screenshots/mobile/error-500.png; PortfolioRoutesTest::test_store_requires_a_skill_and_rejects_unsafe_urls	Verified
H10	Help and documentation	resources/views/help.blade.php; resources/views/layouts/app.blade.php	Link a help page and shortcut dialog from every screen.	docs/screenshots/mobile/help-faq.png; docs/screenshots/mobile/keyboard-shortcuts.png	Verified

Universal Design

Table 21. Universal Design HCI tracker

ID	Principle	Where applied	How applied	Evidence	Status
U1	Equitable use	resources/views/layouts/app.blade.php; resources/css/app.css	Provide keyboard and screen-reader access to core tasks.	tests/e2e/portfolio-accessibility.spec.js::keyboard-first create, publish, manage, template, and delete-undo flows stay accessible; docs/screenshots/mobile/keyboard-shortcuts.png	Verified
U2	Flexibility in use	resources/js/app.js; resources/views/layouts/app.blade.php	Offer shortcuts and ordinary pointer controls.	docs/screenshots/mobile/keyboard-shortcuts.png; docs/screenshots/desktop/keyboard-shortcuts.png	Verified
U3	Simple and intuitive use	resources/views/portfolios/create.blade.php; resources/views/portfolios/select-template.blade.php; resources/views/portfolios/show.blade.php	Label the path from portfolio information through template choice to sharing.	docs/screenshots/mobile/step-1-your-info.png; docs/screenshots/mobile/step-2-choose-template.png	Verified
U4	Perceptible information	resources/css/app.css; resources/views/portfolios/form.blade.php	Use contrast-tested text and pair status colors with readable labels.	ColorContrastTest::test_light_and_dark_design_tokens_meet_wcag_contrast_thresholds; docs/screenshots/mobile/step-3-done-share.png	Verified
U5	Tolerance for error	routes/web.php; resources/views/portfolios/deleted.blade.php	Keep deleted content recoverable for 10 seconds, then permanently remove it.	PortfolioRoutesTest::test_delete_can_be_undone_during_the_ten_second_window; PortfolioModelTest::test_purge_command_removes_portfolios_after_the_undo_period; docs/screenshots/mobile/delete-undo.png	Verified
U6	Low physical effort	resources/css/app.css; resources/js/app.js	Support keyboard navigation and give controls generous spacing.	tests/e2e/portfolio-accessibility.spec.js::keyboard-first create, publish, manage, template, and delete-undo flows stay accessible; docs/screenshots/mobile/keyboard-shortcuts.png	Verified
U7	Size and space for approach and use	resources/css/app.css	Give interactive controls at least 44 CSS pixels and adapt layouts for narrow screens.	tests/e2e/portfolio-accessibility.spec.js::keyboard-first create, publish, manage, template, and delete-undo flows stay accessible; docs/screenshots/mobile/step-1-your-info.png	Verified

Color theory

Table 22. Color theory HCI tracker

ID	Principle	Where applied	How applied	Evidence	Status
C1	Deliberate color scheme per template	resources/css/app.css; docs/DECISIONS.md	Use monochromatic Simple neutrals, analogous blue Modern hues, and complementary plum/magenta with warm cream for Creative.	docs/screenshots/mobile/template-simple.png; docs/screenshots/mobile/template-modern.png; docs/screenshots/mobile/template-creative.png	Verified
C2	Color temperature matches the template's message	resources/css/app.css; docs/DECISIONS.md	Keep Simple restrained, Modern cool and calm, and Creative warm and expressive.	docs/screenshots/mobile/template-simple.png; docs/screenshots/mobile/template-modern.png; docs/screenshots/mobile/template-creative.png	Verified
C3	Sufficient value contrast, WCAG AA	resources/css/app.css; tests/Unit/ColorContrastTest.php	Check text pairs at 4.5:1 and control/focus boundaries at 3:1 in light and dark palettes.	ColorContrastTest::test_light_and_dark_design_tokens_meet_wcag_contrast_thresholds; docs/screenshots/mobile/dark-mode-creative.png	Verified
C4	Never rely on color alone	resources/views/portfolios/form.blade.php; resources/views/portfolios/deleted.blade.php; resources/css/app.css	Pair completion, error, success, and undo states with visible text.	PortfolioRoutesTest::test_store_requires_a_skill_and_rejects_unsafe_urls; docs/screenshots/mobile/error-429.png; docs/screenshots/mobile/step-3-done-share.png	Verified
C5	Define all colors as design tokens in one place	resources/css/app.css	Define app and template palette values in one root token block with dark overrides.	ColorContrastTest::test_light_and_dark_design_tokens_meet_wcag_contrast_thresholds; resources/css/app.css	Verified

Visual design principles

Table 23. Visual design principles HCI tracker

ID	Principle	Where applied	How applied	Evidence	Status
V1	Unity	resources/css/app.css; resources/views/layouts/app.blade.php	Reuse typography, spacing, and component styles throughout the app.	docs/screenshots/mobile/manage.png; docs/screenshots/mobile/template-simple.png	Verified
V2	Gestalt principles	resources/css/app.css; resources/views/portfolios/form.blade.php	Group related fields, repeat section styling, and align content.	docs/screenshots/mobile/step-1-your-info.png; docs/screenshots/mobile/template-creative.png	Verified
V3	Visual hierarchy	resources/css/app.css; resources/views/portfolios/stepper.blade.php	Use heading scale, step state, and spacing to guide the current task.	docs/screenshots/mobile/step-2-choose-template.png	Verified
V4	Balance	resources/css/app.css; resources/views/portfolios/templates/simple.blade.php	Balance primary content, supporting hints, and whitespace at each viewport.	docs/screenshots/mobile/template-simple.png; docs/screenshots/mobile/template-creative.png	Verified
V5	Contrast	resources/css/app.css; tests/Unit/ColorContrastTest.php	Separate text, controls, and focus states with measured contrast.	ColorContrastTest::test_light_and_dark_design_tokens_meet_wcag_contrast_thresholds; docs/screenshots/mobile/dark-mode-creative.png	Verified
V6	Scale	resources/css/app.css; resources/views/portfolios/templates/creative.blade.php	Use a consistent readable type and component scale with responsive headings.	docs/screenshots/mobile/template-creative.png; docs/screenshots/desktop/template-creative.png	Verified
V7	Dominance	resources/css/app.css; resources/views/portfolios/create.blade.php	Make the current heading and primary action the clearest elements.	docs/screenshots/mobile/step-3-done-share.png; docs/screenshots/mobile/step-1-your-info.png	Verified
V8	Similarity	resources/css/app.css; resources/views/layouts/app.blade.php	Style controls with the same role consistently throughout the application.	docs/screenshots/mobile/manage.png; docs/screenshots/mobile/edit.png	Verified

User needs

Table 24. User needs HCI tracker

ID	Principle	Where applied	How applied	Evidence	Status
R1	Documented personas and tasks	docs/USERS_AND_NEEDS.md	Describe student, freelancer, and job seeker contexts and tasks.	HciTrackerTest::test_documented_personas_and_core_tasks_cover_target_users; docs/USERS_AND_NEEDS.md	Verified
R2	Documented assumptions to validate	docs/USERS_AND_NEEDS.md	Separate assumptions from measured user research.	HciTrackerTest::test_user_research_assumptions_are_labeled_as_unvalidated; docs/USERS_AND_NEEDS.md	Verified
R3	Iteration log of changes from evaluation	docs/USERS_AND_NEEDS.md	Record evaluation evidence, findings, design changes, and follow-up work.	HciTrackerTest::test_iteration_log_records_evaluation_findings_and_design_changes; docs/USERS_AND_NEEDS.md; docs/COGNITIVE_WALKTHROUGH.md	Verified